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# FROM COSPRO TO CROSS-FITTED CONCEPT DISTILLATION: A RESEARCH LOG OF THE `CONCEPT-FACTORS` BRANCH

**Spur\_SpLiCE project**

Branch `concept-factors`, 26 September to 2 October 2026

## ABSTRACT

This document records how the `concept-factors` branch left the CoSpRo method and arrived at cross-fitted concept distillation. CoSpRo builds a teacher graph from concept projections of a frozen CLIP/SpLiCE model and lifts Waterbirds worst-group accuracy (WGA) by four points over SimCLR, while on MetaShift it stays below SimCLR. We replaced its prior about which concept is spurious with a symmetric goal: keep strongly correlated concept factors on separate linear directions of the student, so that a group-balanced linear probe can pick the side the task needs. Six rounds of experiments followed. Distilling SpLiCE factors into a SimCLR ResNet doubles WGA on Spur-CIFAR10 from the first attempt and leaves MetaShift unchanged through reweighting, whitening and per-factor contrastive blocks. Shuffled-target controls showed that every per-image auxiliary loss is met by memorization on MetaShift’s 1,700 images; a ridge regression fitted in closed form on one half of each batch (cross-fitting) removed memorization in the head and isolated the concept content as the source of the Spur-CIFAR10 gain. A revised concept grouping, learnability bounds from CLIP features and a hold-out of images from the concept loss then showed that the student memorizes in its backbone on MetaShift and generalizes on Spur-CIFAR10. The configuration that came out of these diagnostics, meaning-grouped factors distilled by the cross-fit at weight 10, reaches 55.3% validation WGA on Waterbirds against 45.8% for SimCLR and 42.8% for its shuffled control, the best Waterbirds result of the project from random initialization. It is now frozen and runs unchanged on CelebA, Spur-CIFAR10 and further Waterbirds seeds.

## 1 OVERVIEW

Table 1 lists the stages of the branch. Each stage answered a question the previous one raised, and most answers were negative. The negative answers matter as much as the final positive one, because each of them removed an explanation for the MetaShift failure and left the next one in view: the choice of target factors, then the loss, then memorization in the head, then the grouping, then the frequency of the factors and finally memorization in the backbone.

Table 1: Stages of the branch. Dates are 2026; WGA is validation worst-group accuracy.

| Date         | Stage                       | Outcome                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------|
| before 26.09 | CoSpRo                      | +3.8 WGA on Waterbirds test; below SimCLR on MetaShift                                               |
| 26.09        | Concept factors, F1 and F2  | F2 doubles Spur-CIFAR10 WGA; MetaShift unchanged                                                     |
| 27.09        | Rounds 1 and 2 on MetaShift | loss weight, whitening, atypicality weights: majority groups rise, minority groups stay near 50      |
| 27–28.09     | Round 3                     | balancing weights and concept blocks (F3); shuffled controls fit as well as real targets             |
| 28.09        | Cross-fitted F2 (xfit)      | first version diverges; the repaired xfit matches F2 on Spur-CIFAR10 and its control falls to SimCLR |
| 28.09        | Concept grouping revisited  | synonyms rarely co-activate; meaning grouping by raw text embeddings                                 |
| 29.09        | Learnability bounds         | rare factors are unpredictable from a half batch; factors from 5%                                    |
| 29–30.09     | Loss weight 3 and 10        | training-image scores exceed CLIP’s three times; WGA flat                                            |
| 30.09        | Unseen-image diagnostic     | MetaShift memorizes, Spur-CIFAR10 generalizes                                                        |
| 01–02.10     | Waterbirds                  | 55.3 WGA against 45.8 for SimCLR; configuration frozen                                               |

## 2 SETTING AND PROTOCOL

All runs share the CoSpRo protocol. A ResNet-18 trains from random initialization with SimCLR (Chen et al., 2020) for 500 epochs (CelebA: 250) at batch size 128, temperature 0.05, learning rate 0.01 and float16 mixed precision on one V100. Every 25 epochs a logistic probe is fitted on a group-balanced training subset and evaluated on the validation split; we report the mean of the last four probes and the standard deviation over seeds unless a table says otherwise. Worst-group accuracy is the lowest accuracy over the groups formed by class and spurious attribute (Sagawa et al., 2020). Concept factors come from SpLiCE (Bhalla et al., 2024) decompositions of OpenCLIP ViT-B/32 embeddings (Radford et al., 2021) over the LAION (10,000 words) or Open Images V7 (20,931 words) vocabularies. Pretraining uses no class or group label.

Table 2: Datasets. MetaShift and Spur-CIFAR10 are development datasets, on which settings are chosen; Waterbirds and CelebA are held out for the final evaluation.

| Dataset                                       | Train images | Spurious structure                                                                  |
|-----------------------------------------------|--------------|-------------------------------------------------------------------------------------|
| MetaShift cats and dogs (Liang and Zou, 2022) | 1,700        | cats indoors and dogs outdoors in 88% of the images; 72 validation images per group |
| Spur-CIFAR10                                  | 45,000       | CIFAR-10 classes with a class-correlated coloured line; 100 groups                  |
| Waterbirds (Sagawa et al., 2020)              | 4,795        | bird type correlated with land or water background                                  |
| CelebA (Liu et al., 2015)                     | 162,770      | hair colour correlated with gender                                                  |

Two protocol rules shaped the decisions below. The development datasets carry all tuning; Waterbirds and CelebA receive frozen configurations, so the label-free claim holds for them. MetaShift WGA is noisy: SimCLR over ten seeds scores  $44.6 \pm 6.9$ , so two-seed differences below five points carry little evidence and we leaned on mechanistic diagnostics more than on WGA there.

## 3 STARTING POINT: COSPRO

CoSpRo removes the subspace of one concept group from CLIP embeddings, keeps image pairs that become closer, filters them by activation contrast, residual concept agreement and synthetic null groups and transfers the resulting sparse graph to a SimCLR student through graph-aware batches and a relational KL loss. On Waterbirds it reaches  $50.64 \pm 1.97$  test WGA against  $46.82 \pm 3.65$  for SimCLR over four seeds. Two observations motivated the branch.

**The gain does not transfer.** On MetaShift, ten seeds give  $42.9 \pm 2.7$  WGA for CoSpRo against  $44.6 \pm 6.9$  for SimCLR, and the LAION variant reaches  $47.7 \pm 4.3$ , within the SimCLR spread. On Spur-CIFAR10 one seed gives 19.6 against 15.6.

**The method carries a prior.** Projection relations help when removing a concept reveals a same-class pair across contexts, which assumes that the spurious attribute is a removable context concept. Label-free alternatives carry other priors: LateTVG (Hamidieh et al., 2024) and learning-speed methods (Zhu et al., 2023) assume that the spurious feature is the easy one. The CoSpRo graph also contains wrong-class edges, and background stays linearly decodable at 91%.

## 4 CONCEPT FACTORS: THE IDEA

Deep feature reweighting (Kirichenko et al., 2023) shows that a linear head retrained on group-balanced data recovers worst-group accuracy whenever the representation encodes the core feature apart from the spurious one. Our probe is exactly such a head. The SSL failure to prevent is therefore the fusion of two correlated features into one direction, or the suppression of the harder feature by the easier one, which contrastive losses are known to produce (Chen et al., 2021; Robinson et al., 2021). Keeping every strongly correlated pair of concepts apart serves both sides of the pair, and the balanced probe chooses the side the task needs. No concept is declared spurious.

SpLiCE supplies the ingredients without labels. A *factor* is a concept group of the CoSpRo grouping stage; its value on image  $i$  is the summed SpLiCE code  $a_{ik} = \sum_{c \in G_k} c_{ic}$ . Factors outside a frequency band (2% to 90% of the images by default) are dropped. Two mechanisms used them.

**F1, concept-conditioned batches.** A share of the batches holds only images that show one factor of a strongly correlated pair (phi coefficient of presences). Inside such a batch the factor tells no image apart, so InfoNCE has to separate the images by everything else, including the partner factor. This is conditional contrastive learning (Tsai et al., 2021) with a discovered factor in place of an annotated attribute.

**F2, factor distillation.** A linear head  $W$  on the backbone feature  $h_i$  predicts the factor targets  $t_i$  of both views,

$$\mathcal{L} = \mathcal{L}_{\text{SimCLR}} + \lambda \frac{1}{2B} \sum_{i,v} \|Wh_i^{(v)} - t_i\|^2 / K. \quad (1)$$

The targets are standardized factor values or their ZCA-whitened version (Kessy et al., 2018), which leaves each factor the part the other factors do not explain and so is large exactly on the images that break a correlation. The head is linear so that the factors must be linearly readable, as the probe reads them. A *shuffled* control assigns each image the targets of another image: same statistics, no image content.

## 5 ROUNDS 1 TO 3: DISTILLATION HELPS WHERE DATA IS PLENTIFUL

### 5.1 ROUND 1

The first inspection of the factors showed two defects, both fixed before training. Taking the 64 most frequent groups dropped the scene concepts of MetaShift and most line colours of Spur-CIFAR10, and the top correlated pairs were two names for one thing (cat breeds, lorry and truck). We kept every group in the band and merged groups whose CLIP image responses correlate above 0.9.

On Spur-CIFAR10 the gain was large and content-specific: real targets beat the shuffled control by nine points of WGA and eleven of accuracy (Table 3). The control itself added nine points over SimCLR at equal accuracy, so the auxiliary regression also helps as such; Section 6 removes this ambiguity. On MetaShift the head explained 12% of the factor variance and the arms differed by less than the seed noise. We counted Spur-CIFAR10 as a success and MetaShift as an unexplained failure.

Table 3: Round 1, seeds 1 and 2. EV is the explained variance of the targets by the trained head.

| Dataset      | Arm              | WGA            | Acc. | EV   |
|--------------|------------------|----------------|------|------|
| Spur-CIFAR10 | SimCLR           | $15.6 \pm 1.0$ | 59.5 |      |
|              | F2, standardized | $32.9 \pm 1.6$ | 69.5 | 0.62 |
|              | F2, shuffled     | $24.2 \pm 0.4$ | 58.6 | 0.02 |
| MetaShift    | SimCLR           | $47.7 \pm 4.2$ | 55.8 |      |
|              | F2, standardized | $43.2 \pm 1.2$ | 58.2 | 0.12 |
|              | F2, shuffled     | $42.2 \pm 4.7$ | 51.0 | 0.06 |

## 5.2 ROUND 2: POINTING THE LOSS AT THE MINORITY IMAGES

Every MetaShift arm with real targets raised average accuracy, while the minority groups (cats outdoors and dogs indoors) stayed near 45 to 55%. Our reading was that standardized targets reproduce the dataset’s correlations, and several Open Images concepts fuse class and context (“Cat bed”, “Dog walking”). Round 2 therefore tested ways to point F2 at the images that break the correlations without labels: loss weights 3 and 10, whitened targets and per-image *atypicality* weights (the mean squared whitened factor value).

Table 4: MetaShift, seeds 1 to 4 unless noted. Group accuracies in the order cat-indoor, cat-outdoor, dog-indoor, dog-outdoor.

| Arm                        | WGA            | Acc. | Groups            | EV   |
|----------------------------|----------------|------|-------------------|------|
| SimCLR                     | $44.8 \pm 9.3$ | 54.8 | 55 / 49 / 53 / 62 |      |
| F2, weight 3               | $46.6 \pm 5.7$ | 61.5 | 70 / 47 / 54 / 74 | 0.41 |
| F2, weight 3, shuffled     | $43.3 \pm 8.1$ | 54.8 | 57 / 45 / 53 / 65 | 0.33 |
| F2, weight 10 (seeds 1, 2) | $47.0 \pm 7.1$ | 59.7 | 62 / 46 / 53 / 75 | 0.85 |
| F2, whitened, weight 3     | $47.9 \pm 6.4$ | 56.9 | 57 / 51 / 55 / 65 | 0.31 |
| F2, weight 3, atypicality  | $50.5 \pm 3.8$ | 58.8 | 59 / 52 / 55 / 70 | 0.61 |

Whitened targets and atypicality weights narrowed the gap between groups by lowering the majority gain; none of them lifted a minority group above the SimCLR spread (Table 4). We read this as the regression learning through the co-occurrence itself: in 88% of the images the indoor features predict the cat factors, so one fused “cat and indoor” direction serves the loss.

## 5.3 ROUND 3: MAKING THE CO-OCCURRENCE USELESS

Two mechanisms targeted the fused direction directly. *Balancing weights* minimize the mean squared correlation of the factor presences under per-image weights with an entropy penalty, so a co-occurrence stops paying off in the weighted regression; this is sample reweighting for independence as in stable learning (Zhang et al., 2021). *F3 concept blocks* give every factor its own small linear block and apply a supervised contrastive loss (Khosla et al., 2020) inside the block, with images that share the factor as positives and a pair weight that grows with how much the two images’ other factors differ. A fused direction cannot separate “cat on sofa” from “dog on sofa” inside the cat block, so the backbone needs a cat direction that survives the context; pairs that share one factor and differ elsewhere identify that factor in the weak-supervision sense of Locatello et al. (2020) and Yao et al. (2024).

Table 5: Round 3 on MetaShift, seeds 1 to 4.

| Arm                             | WGA            | Acc. | Groups            |
|---------------------------------|----------------|------|-------------------|
| F2, weight 3, balancing weights | $44.2 \pm 8.1$ | 59.0 | 64 / 44 / 55 / 72 |
| F3 concept blocks               | $43.8 \pm 4.1$ | 60.0 | 69 / 45 / 51 / 75 |
| F3, all positive pairs alike    | $49.7 \pm 2.5$ | 62.6 | 73 / 50 / 52 / 75 |
| F3, shuffled concept sets       | $46.4 \pm 3.5$ | 56.2 | 56 / 48 / 52 / 68 |
| F3 and balanced F2              | $49.4 \pm 3.8$ | 64.4 | 74 / 52 / 55 / 78 |

No arm lifted a minority group (Table 5). The controls explained why. F2 at weight 3 explained 0.41 of the real target variance and 0.33 of the shuffled one, and the block loss ended at 3.34 with real concepts and 3.37 with shuffled ones. On 1,700 images a network can fit per-image targets through the features SimCLR learns anyway to tell images apart, and random targets are fitted almost as well as real ones, as Zhang et al. (2017) and Arpit et al. (2017) showed for labels. Any auxiliary loss attached to single images is then met without learning concepts. This was the first mechanistic finding of the branch, and it redirected the work from the target design to the estimator.

## 6 CROSS-FITTING: NO MEMORIZATION IN THE HEAD

The cross-fitted F2 (xfit) replaces the trained head by a ridge regression fitted in closed form on one half  $A$  of the batch and evaluated on the other half  $B$ , then the reverse. With unit-norm, centred features  $H$ , targets  $T$  and per-image weights  $D$ , the kernel form

$$\hat{T}_B = H_B H_A^\top (D_A H_A H_A^\top + \rho I)^{-1} D_A T_A \quad (2)$$

needs one  $n \times n$  solve and is differentiable, as in the meta-learning heads R2D2 (Bertinetto et al., 2019) and MetaOptNet (Lee et al., 2019). The loss is the held-out error on  $B$ . Both views of an image stay in one half. A feature that only identifies an image carries nothing from  $A$  to  $B$ , so only concept directions shared across images lower the loss; the construction is the sample splitting of double machine learning (Chernozhukov et al., 2018) applied to a training loss. The held-out explained variance of the batch fit is logged as a memorization-free diagnostic.

**First version: failure.** The first runs produced NaN features on three of four Spur-CIFAR10 runs and negative held-out explained variance everywhere. The ridge scaled with the feature norm and was small, so fits of 64 images in 512 dimensions nearly interpolated and the solve became ill conditioned. Scaling every feature row to unit norm and using an absolute ridge  $\rho = 1$  keeps the eigenvalues of the system at least  $\rho$  (Hoerl and Kennard, 1970).

Table 6: Repaired cross-fit, seeds 1 and 2. Batch EV is the held-out explained variance of the batch fit at epoch 500.

| Dataset      | Arm                     | WGA            | Acc. | Batch EV |
|--------------|-------------------------|----------------|------|----------|
| Spur-CIFAR10 | SimCLR                  | $15.6 \pm 1.0$ | 59.5 |          |
|              | F2, standardized        | $32.9 \pm 1.6$ | 69.5 | 0.10     |
|              | xfit                    | $33.6 \pm 2.3$ | 69.7 | 0.13     |
|              | xfit, shuffled          | $18.2 \pm 3.2$ | 57.5 | -0.02    |
| MetaShift    | xfit                    | $43.9 \pm 3.2$ | 54.8 | -0.01    |
|              | xfit, shuffled          | $41.7 \pm 0.0$ | 54.4 | -0.03    |
|              | xfit, ridge 10          | $42.5 \pm 2.2$ | 54.4 | -0.01    |
|              | xfit, weight 3          | $43.9 \pm 0.7$ | 57.8 | 0.00     |
|              | xfit, balancing weights | $46.9 \pm 4.4$ | 58.5 | -0.01    |

On Spur-CIFAR10 the cross-fit matches F2, and its shuffled control falls to the SimCLR level (Table 6). Where the shuffled F2 had kept nine points, the cross-fitted control keeps 2.6, so the whole remaining gain comes from the concept content. We counted this as the first clean attribution of the branch. On MetaShift the 145 factors stayed at the noise level under every setting, which pointed away from the loss and toward the factors themselves.

## 7 CONCEPT GROUPING REVISITED

Inspecting single images showed that the factors were rarely groups at all. On Waterbirds with Open Images at the default thresholds, 499 of 502 groups were single concepts, and the strongest concepts of an image often fell below the frequency cut (“Footpath” on 0.94% of the images). We measured four candidate similarity signals on concept pairs of the Waterbirds LAION cache (Table 7).

Two properties of SpLiCE explain the failure of the original rule, which links two concepts only when both their centred text similarity and their co-activation pass a threshold. The L1 penalty selects

Table 7: Similarity of concept pairs on Waterbirds (LAION). Text: cosine of the centred SpLiCE dictionary directions; raw text: cosine of the uncentred CLIP text embeddings; co-activation: cosine of the code columns; response: correlation of the images’ CLIP alignment with the two concepts.

| Pair             | Text | Raw text | Co-activation | Response |
|------------------|------|----------|---------------|----------|
| path, walkway    | 0.23 | 0.85     | 0.37          | 0.66     |
| heron, egret     | 0.70 | 0.87     | 0.09          | 0.79     |
| ocean, sea       | 0.71 | 0.94     | 0.00          | 0.94     |
| forest, woodland | 0.49 | 0.90     | 0.08          | 0.89     |
| bamboo, jungle   | 0.24 | 0.67     | 0.10          | 0.67     |
| gull, seascape   | 0.16 | 0.65     | 0.02          | 0.58     |
| heron, marsh     | 0.08 | 0.67     | 0.06          | 0.11     |

one of several correlated dictionary atoms, the known behaviour of the lasso on correlated predictors (Tibshirani, 1996; Zou and Hastie, 2005), so synonyms rarely fire on the same image. Centring the dictionary removes the shared component of related words and lowers their cosine. Lowering both thresholds then joins concepts through chains of pairwise links, the single-linkage behaviour of connected components: at text 0.3 and co-activation 0.1 one group held hummingbird, parrot, jungle and bamboo, so the factor itself fused a bird with its background.

The *meaning grouping* clusters concepts by average linkage (Sokal and Michener, 1958) on the raw CLIP text embeddings, where synonyms reach 0.85 to 0.94 and object–context pairs 0.64 to 0.72. Average linkage joins two clusters only when their members agree on average, which stops the chains. An optional veto drops pairs whose image responses correlate below 0.5. Concepts from 0.2% frequency enter the grouping, so rare synonyms join before the factor band applies to their sum. A sweep over 576 factor sets on MetaShift and Waterbirds chose text 0.85 and response 0.5 on MetaShift, which yields groups such as {kitty, kitten, cat}, {canine, dog} and {cozy, comfy, cosy} with LAION. Open Images groups also join homonyms and similar Latin names (Quail and Quail meat).

The same sweep put pair precision, the share of the top correlated pairs that join a class factor to an attribute factor, at zero for both grouping methods. The phi coefficient ranks within-scene pairs first (windshield and driving 0.64, toilet and sink 0.57) and class–context pairs last (dog and lawn 0.18, cat and couch 0.04), because phi penalizes pairs of very different frequency (cat on 36% of the images, couch on 3%). We set F1 aside: F2 and xfit use every factor and need no pairs.

## 8 LEARNABILITY: WHAT THE STUDENT CAN AND DOES LEARN

### 8.1 A BOUND FROM CLIP FEATURES

The cross-fit can only reward what a ridge fitted on 64 images can predict. We therefore ran the same fit on frozen CLIP image embeddings, the features the targets come from, as an upper bound (Table 8). With the 278 meaning factors from 1%, even CLIP scores below zero: a factor on 1% of the images occurs 0.6 times per half batch, and the loss consists of noise. Restricting the factors to those on at least 5% of the images (24 factors) raises the bound to 0.16 against  $-0.07$  for shuffled targets.

Table 8: Held-out explained variance of a ridge ( $\rho = 1$ ) fitted on  $n$  MetaShift images and evaluated on  $n$  others, averaged over 40 draws.

| Factors       | $n$ | CLIP features | Shuffled targets |
|---------------|-----|---------------|------------------|
| 278 (from 1%) | 64  | $-0.015$      | $-0.079$         |
| 278 (from 1%) | 800 | 0.140         | $-0.086$         |
| 24 (from 5%)  | 64  | 0.161         | $-0.069$         |
| 11 (from 10%) | 64  | 0.218         | $-0.061$         |

With 24 factors and weight 1 the student’s batch score rose to 0.05 at epoch 500 and was still rising, against  $-0.02$  for the control, while WGA stayed at  $46.0 \pm 2.2$  against  $46.7 \pm 1.7$ . We read the rising score as concepts being learned and raised the weight.

## 8.2 WEIGHT 3 AND 10: SCORES ABOVE THE BOUND

Every concept-factor run now writes a per-factor record: a five-fold ridge ( $\rho = 10$ ) from the training features to each real standardized factor, beside the score of the same fit on CLIP features. Shuffled arms score the real factors too, so they show what SimCLR learns of the concepts by itself.

Table 9: MetaShift, 24 meaning factors, seeds 1 and 2. Train EV is the five-fold score on the training images; CLIP features score 0.26.

| Arm                      | Train EV | WGA  | Acc. | Groups            |
|--------------------------|----------|------|------|-------------------|
| xfit, weight 3, shuffled | 0.02     | 47.6 | 58.1 | 68 / 53 / 50 / 59 |
| xfit, weight 3           | 0.33     | 43.2 | 59.6 | 74 / 45 / 48 / 69 |
| xfit, weight 10          | 0.79     | 45.5 | 61.0 | 74 / 51 / 48 / 71 |

At weight 10 every factor reached 0.72 to 0.87 on the training images, three times the CLIP score and alike for factors CLIP explains well (cat, 0.68) and badly (helper, 0.14). A logistic probe on the 24 factor values themselves reaches 77% WGA and 89% accuracy on group-balanced splits of the training images, and CLIP embeddings reach 87% and 92%; the student stays at 45% and 61% on validation. The factors carry the information the probe needs; the student holds something else.

## 8.3 UNSEEN IMAGES: MEMORIZATION IN THE BACKBONE

Cross-fitting holds images out of the ridge and leaves the backbone free to place each training image’s target in its own features, where one shared linear map reads it back; the whole-dataset score cannot see this. We kept a fixed fifth of the training images out of the concept loss (SimCLR still trains on them) and scored the factors on those images with a ridge fitted on the others.

Table 10: Explained variance of the factors on the images in the concept loss (seen) and on the fifth held out from it (unseen), epoch 500, seeds 1 and 2.

| Dataset      | Arm             | Seen | Unseen | CLIP unseen | WGA  | Acc. |
|--------------|-----------------|------|--------|-------------|------|------|
| MetaShift    | xfit, weight 10 | 0.60 | 0.04   | 0.26        | 47.9 | 60.4 |
|              | shuffled        | 0.01 | 0.02   | 0.26        | 43.8 | 52.6 |
| Spur-CIFAR10 | xfit            | 0.31 | 0.27   | 0.42        | 30.2 | 68.7 |
|              | shuffled        | 0.10 | 0.10   | 0.42        | 21.7 | 58.1 |

Table 10 separates the two datasets. On Spur-CIFAR10 the student carries the factors to new images at two thirds of the CLIP score, for the classes (aircraft 0.75, horse 0.74, truck 0.72) and for the line colour (purple 0.62) alike, the separate class and attribute directions that deep feature reweighting needs. On MetaShift the unseen score peaks near 0.08 at epochs 100 to 200 and falls to 0.04 while the seen score climbs to 0.60; the class factors carry over in part (cat 0.37, dog 0.23) and the context factors do not (cozy 0.03). The student memorizes the targets of MetaShift’s 1,700 images and generalizes them on the 45,000 images of Spur-CIFAR10. This explains why the majority groups of MetaShift rose in every round while the minority groups stayed flat.

## 9 WATERBIRDS

Waterbirds received frozen configurations throughout. F2 as frozen after round 1 lowered WGA to  $41.6 \pm 6.1$  against  $45.8 \pm 1.7$  for SimCLR over four seeds. The repaired cross-fit with the default 217 Open Images factors learned little and memorized nothing (Table 11): seen and unseen scores agree and exceed the control by a few hundredths, and validation accuracy stays at chance.

The configuration that came out of the MetaShift diagnostics, meaning groups at text 0.85 and response 0.5, factors from 5% and weight 10, lifts every Waterbirds group, minority groups included,

Table 11: Waterbirds validation. SimCLR: seeds 1 to 4; other arms: seeds 1 and 2 with a fifth of the images held out of the concept loss; the meaning arms report the final probe.

| Arm                                 | Seen | Unseen | WGA            | Acc.        | Groups            |
|-------------------------------------|------|--------|----------------|-------------|-------------------|
| SimCLR                              |      |        | $45.8 \pm 1.7$ | 52.3        | 47 / 54 / 52 / 64 |
| xfit, 217 factors, weight 1         | 0.04 | 0.03   | 46.5           | 51.1        |                   |
| shuffled                            | 0.02 | 0.02   | 43.9           | 50.5        |                   |
| xfit, 70 meaning factors, weight 10 | 0.30 | 0.09   | <b>55.3</b>    | <b>59.2</b> | 56 / 59 / 66 / 68 |
| shuffled                            | 0.01 | 0.02   | 42.8           | 52.6        | 43 / 57 / 58 / 66 |

to 55.3% WGA (57.8 and 52.7 for the two seeds). This exceeds SimCLR by 9.5 points, its shuffled control by 12.5 points and the CoSpRo validation result ( $50.0 \pm 3.2$ ) by 5.3 points. The student stores part of the targets (seen 0.30) and carries a third of the CLIP score to unseen images (0.09 against 0.02 for the control), more than twice the MetaShift value. Waterbirds, with 4,795 images, sits between the memorizing and the generalizing dataset.

We count this as the first success of the branch on a dataset where CoSpRo already worked, with two caveats: two seeds, and the hold-out of a fifth of the images. Its settings came from MetaShift diagnostics and the learnability bounds; the weight of 10 was chosen to probe memorization on MetaShift, and no setting was tuned on Waterbirds labels. The configuration is frozen from here on.

## 10 WHERE WE STAND AND HOW WE CONTINUE

### Conclusions so far.

1. Distilling SpLiCE concept factors into a SimCLR student improves worst-group accuracy when the student can generalize the factors: Spur-CIFAR10 (WGA 15.6 to 33.6) and Waterbirds (45.8 to 55.3).
2. Cross-fitting makes the attribution clean: shuffled targets lose the gain, so the concept content carries it.
3. On small datasets an auxiliary per-image target is memorized. Cross-fitting removes memorization in the head; the backbone still memorizes, and only a score on images outside the concept loss shows it.
4. The grouping must join synonyms that SpLiCE never activates together and must avoid chains through context; the meaning grouping does both. Factors must be frequent enough for a half batch to contain them.
5. Pair selection by phi favours within-scene pairs, so conditioned batches (F1) need another pair statistic before they can work.

**Running and planned experiments.** CelebA trains SimCLR, the weight-1 cross-fit and its control for 250 epochs on 162,770 images; at epoch 188 the cross-fit scores 0.20 on the training factors against 0.07 for the control and 0.28 for CLIP. Waterbirds runs the frozen configuration on seeds 3 and 4 with the hold-out and on seeds 1 to 4 without it, which gives the headline number. Spur-CIFAR10 runs the same configuration to test that one setting serves all datasets, and CelebA follows with it once the weight-1 runs finish. At most eight jobs run at a time.

**Open questions.** Why does Waterbirds generalize part of the factors where MetaShift does not, beyond its threefold size? The 70 Open Images factors include bird families and backgrounds with high CLIP scores (Water bird 0.61, Bamboo 0.76), while the MetaShift factors are dominated by pet-photo vocabulary. Can memorization be made unprofitable on small datasets, for example by mixing images and their targets as in mixup (Zhang et al., 2018), so that storing per-image targets stops lowering the loss? Does a pair statistic that measures how much a context raises the probability of a class revive F1?

**Threats to validity.** Two seeds per arm leave wide intervals, and MetaShift’s 72-image validation groups make its WGA noisy. All values are validation results; the locked test split stays unused until



the final configuration is fixed. Waterbirds served as a diagnostic dataset for the weight-1 cross-fit before the configuration was frozen, so its role as a held-out dataset rests on the absence of any label-based selection on it.

## A OPTIONS AND ARMS

Table 12: Training options of the concept-factor method (`-splice_mode concept_factors`).

| Option                                        | Meaning                                                       |
|-----------------------------------------------|---------------------------------------------------------------|
| <code>-factor_distill_weight</code>           | F2 or xfit loss weight $\lambda$                              |
| <code>-factor_targets</code>                  | standardized, whitened, presence or shuffled                  |
| <code>-factor_cross_fit, -factor_ridge</code> | cross-fitted ridge and its ridge $\rho$                       |
| <code>-factor_sample_weighting</code>         | none, atypicality or balanced                                 |
| <code>-factor_min_frequency</code>            | lowest factor frequency                                       |
| <code>-factor_merge_similarity</code>         | merge groups by image response; 0 disables                    |
| <code>-factor_concept_groups</code>           | groups file, e.g. meaning groups from<br>build_meaning_groups |
| <code>-factor_holdout_fraction</code>         | share of images kept out of the concept loss                  |
| <code>-factor_condition_fraction</code>       | F1 share of conditioned batches                               |
| <code>-factor_block_weight</code>             | F3 concept-block loss weight                                  |

Run records live under `outputs/seeds/factors_<dataset>/`, study summaries under `outputs/reports/factors_<dataset>/summary.md` and per-factor learnability records in `factor_learnability_epoch_<E>.json` beside each run’s status file. The arms are defined in `scripts/submit_factor_study.sh`; the running notes are in `docs/notes/CONCEPT_FACTORS_2026_09_26.md`.

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